

RAK14003 WisBlock LED Bar Graph Module Datasheet

Overview

Description

RAK14003 is an LED Bar Graph that is part of WisBlock Display modules. It consists of 10 configurable LEDs (5 green color LEDs, 3 yellow color LEDs, and 2 red color LEDs). RAK14003 uses the **MCP23017** from Microchip as an I/O Expander and **KEM-102510A-RYG** from Hongke Lighting as the LED bar. Each LED in the module can be controlled separately so the module can build a multipurpose graphic feedback display.

Features

- LED Bar Graph module
- 5 green color LEDs, 3 yellow color LEDs, and 2 red color LEDs
- Each LED can be controlled separately
- 3.3 V Power supply
- Chipset: Microchip MCP23017, Hongke Lighting KEM-102510A-RYG
- Module size: 25 × 45 mm

Specifications

Overview

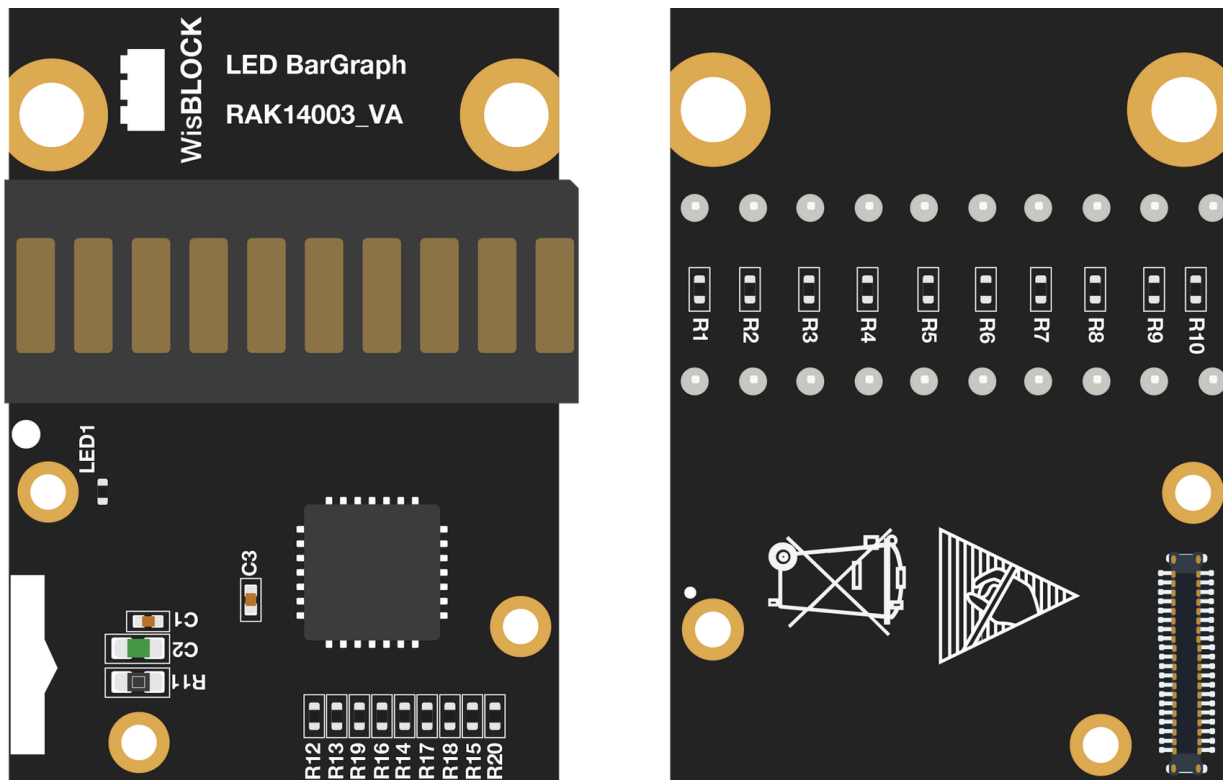


Figure 1: RAK14003 Front and Back View

Mounting

The RAK14003 module can be mounted on the IO slot of the WisBlock Base board. **Figure 2** shows the mounting mechanism of the RAK14003 on a WisBlock Base module, such as the RAK5005-O.

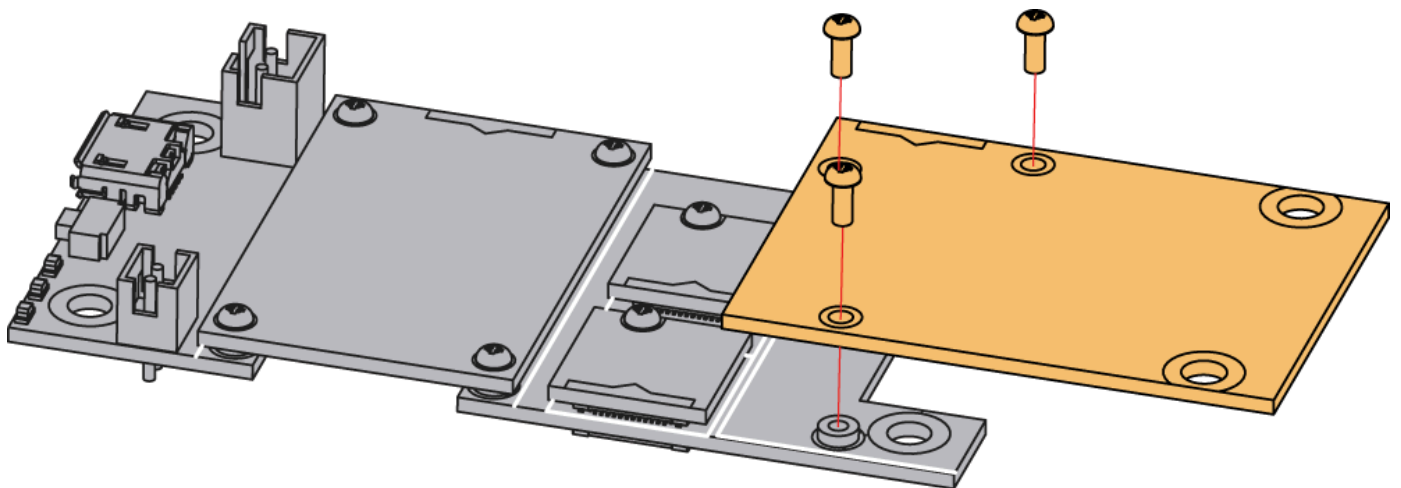


Figure 2: RAK14003 Mounting Mechanism on a WisBlock Base Module

Hardware

The hardware specification is categorized into four parts. It discusses the pinouts and its corresponding functions and diagrams of the module. It also covers the electrical and mechanical characteristics that include the tabular data of the functionalities and standard values of the RAK14003 WisBlock LED Bar Graph Module.

Chipset

Vendor	Part number
Microchip, Hongke Lighting	MCP23017, KEM-102510A-RYG

Pin Definition

The RAK14003 module has a 40-pin WisConnector that is compatible to the WisBlock Base IO Slot. The pin order of the connector and the pinout definition is shown in **Figure 3**.

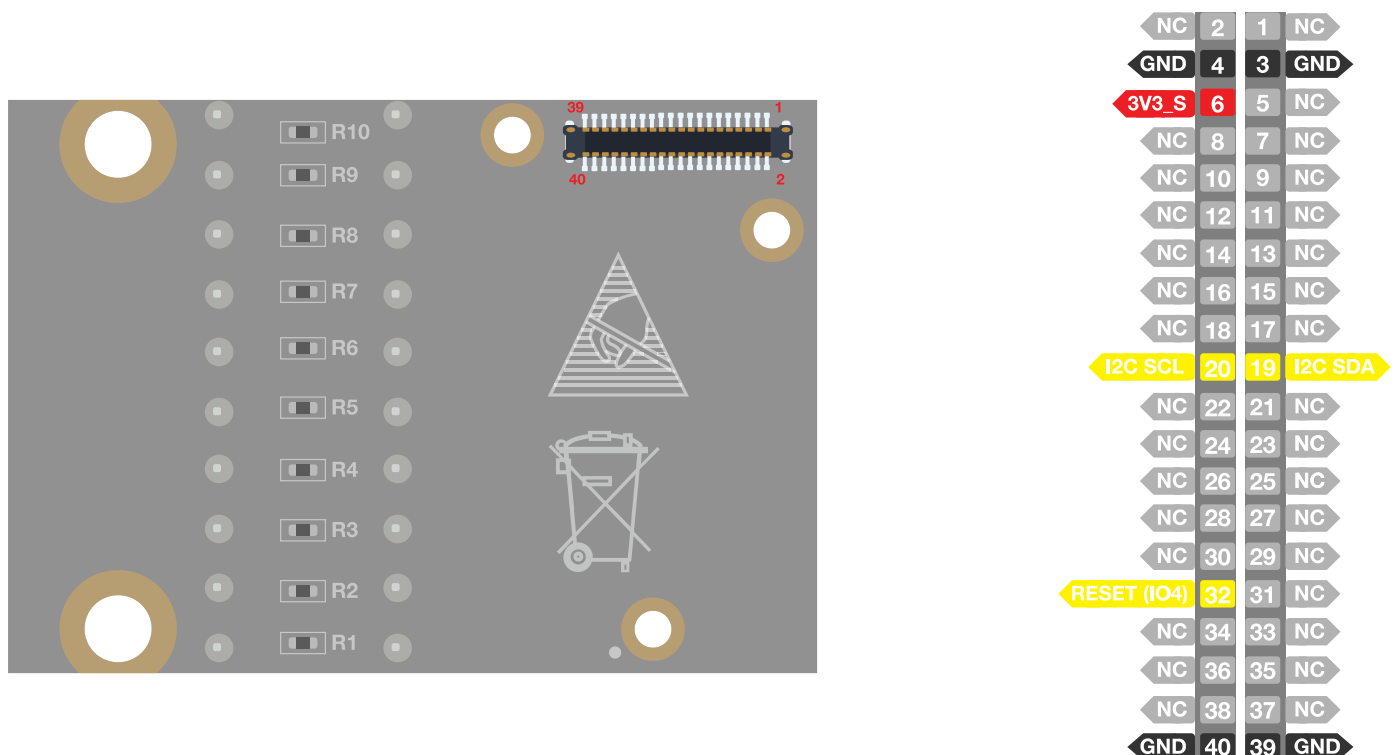


Figure 3: RAK14003 Pinout Schematic

NOTE

- Only I2C related pin, **RESET**, **3V3_S**, and **GND** are connected to the WisConnector.

- **3V3_S** voltage output from the WisBlock Base that powers the RAK14003 module can be controlled by the WisBlock Core via WB_IO2 (WisBlock IO2 pin). This makes the module ideal for low-power IoT projects since the WisBlock Core can totally disconnect the power of the RAK14003 module.

Electrical Characteristics

This section shows the maximum and minimum ratings of the RAK14003 module and its recommended operating conditions. Refer to the table presented below.

Absolute Maximum Ratings

Parameter	Description	Min.	Nom.	Max.	Unit
3V3_S	Power Voltage Range	-0.3	-	5.5	V

Power Supply Ratings

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
3V3_S	Supply Voltage	Input voltage must within this range	-	3.3	-	V
I _{max}	Max Current	Turn ON all LEDs	-	-	170	mA

LED Parameters

Table below shows the LED parameters. Color Code & Chip Characteristic: (Test Condition: IF=20 mA)

Emitting Color	Peak Wavelength		Forward Voltage		Luminous Intensity IF=20 mA (mcd)
	(λp)	Δλ	Typ	Max	Max
Amber	605-615	30	1.9	2.4	30
Super Green	565-575	30	1.9	2.4	80
Super Red	625-640	20	1.8	2.3	100
High Red	640-660	20	1.8	2.3	12
Yellow Green	570-580	30	2.0	2.5	45
Yellow	590-600	35	2.0	2.5	60
Blue	465-475	40	3.0	3.5	200
Pure Green	515-525	40	3.0	3.5	600
White			3.0	3.5	180

Mechanical Characteristics

Board Dimensions

Figure 4 shows the mechanical dimensions of the RAK14003 Module.

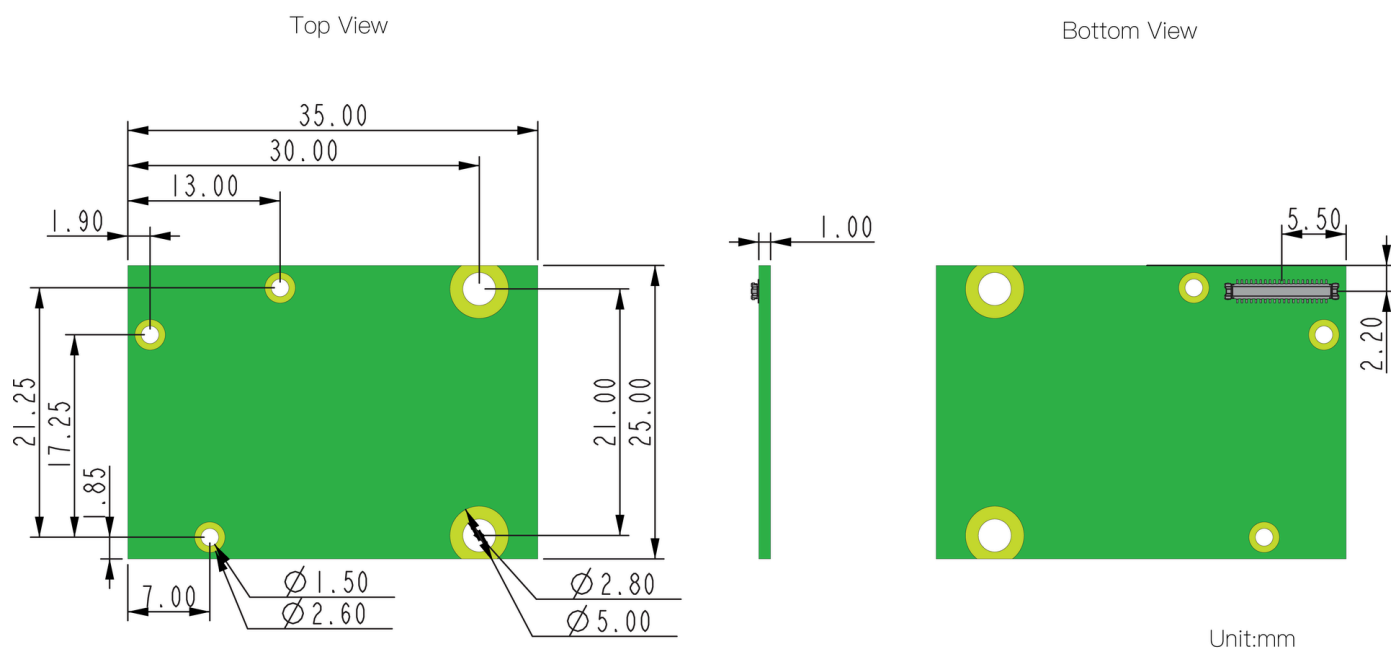


Figure 4: RAK14003 Mechanical Dimensions

WisConnector PCB Layout

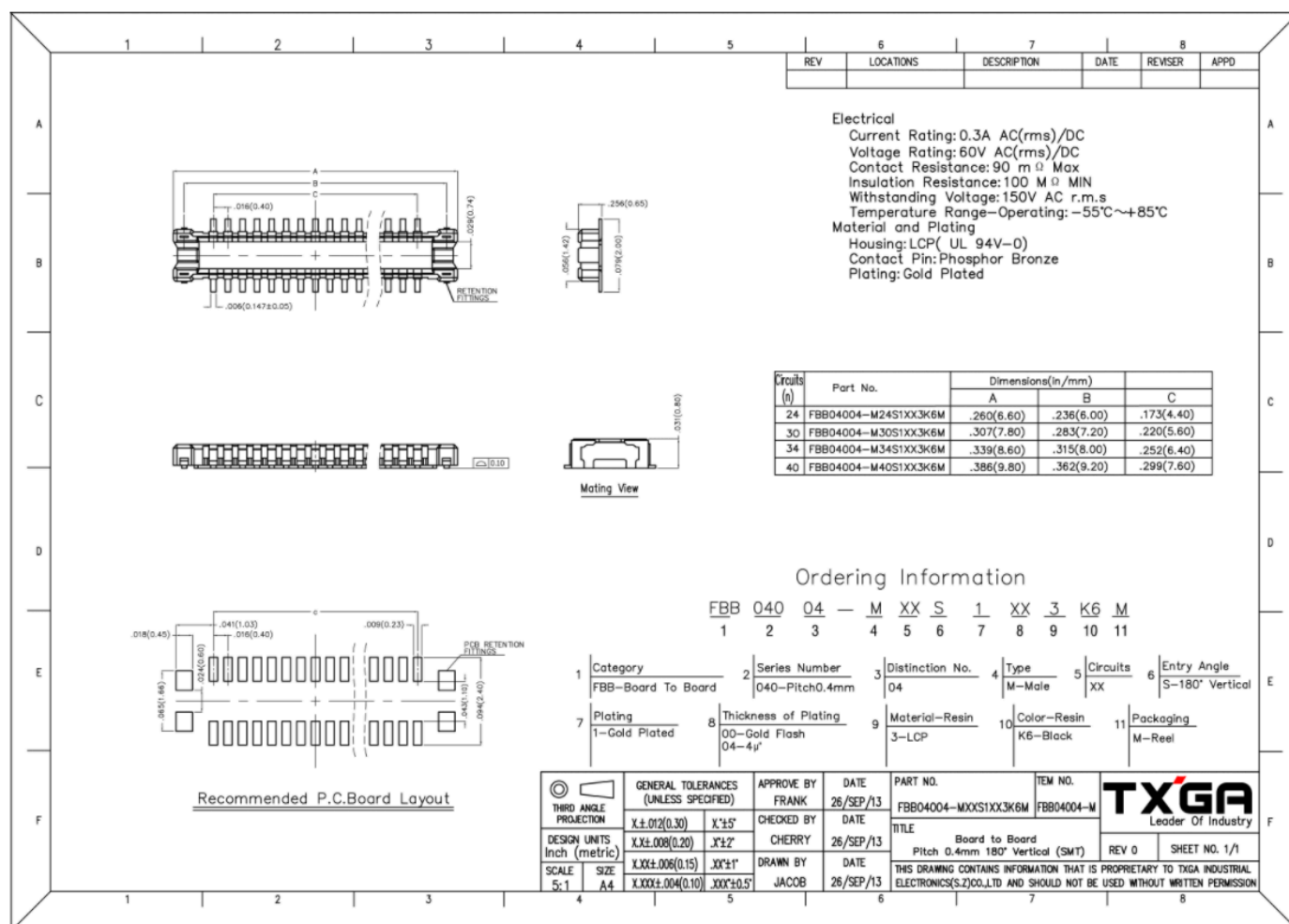


Figure 5: WisConnector PCB Footprint and Recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK14003 module.

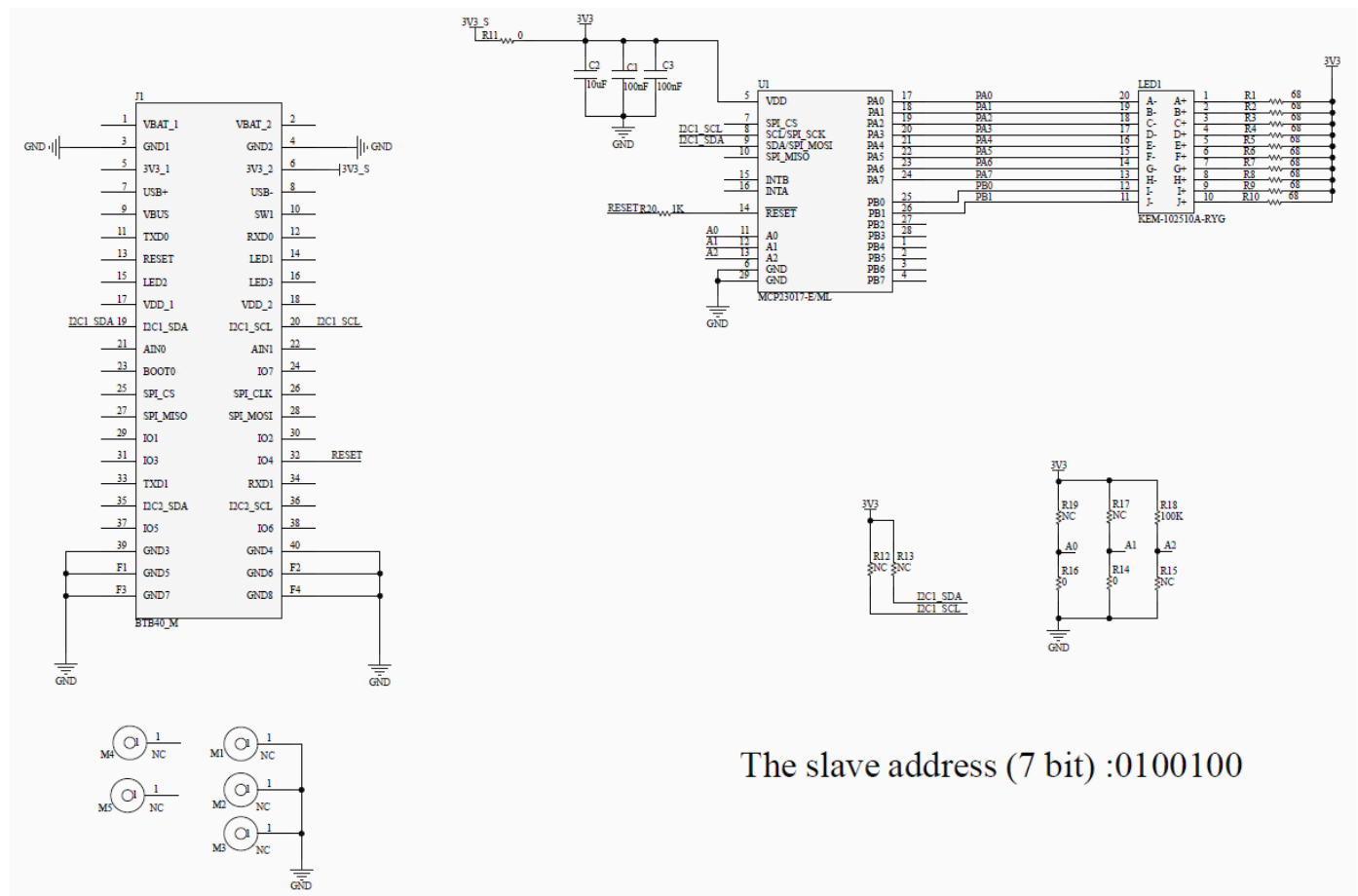


Figure 6: RAK14003 WisBlock Module Schematics

I/O Expander and LED Bar

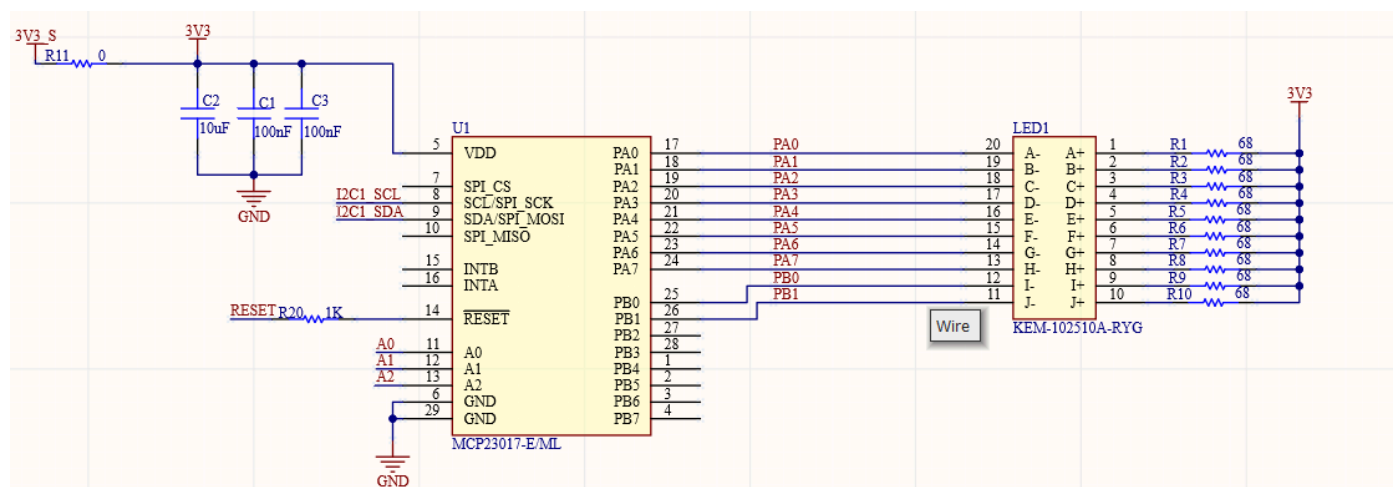


Figure 7: RAK14003 I/O Expander and LED Bar

NOTE

- The current-limiting resistances are 68 Ω .

Device Address Select

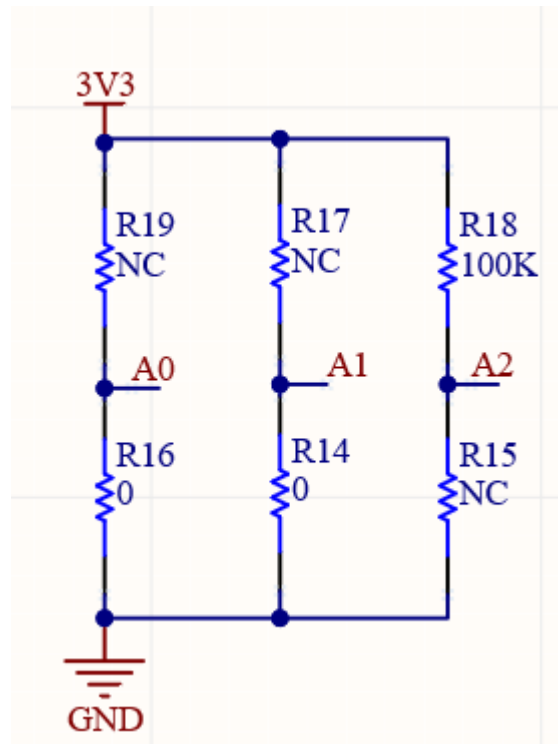


Figure 8: I2C Address Select

NOTE

- You can change the I2C slave address. The default 7-bit I2C address is **0x24**.

I2C Pull-up Resistance

The I2C Pull-up resistors already exist on the WisBlock Base, such as the RAK5005-O, not on the RAK14003 module.

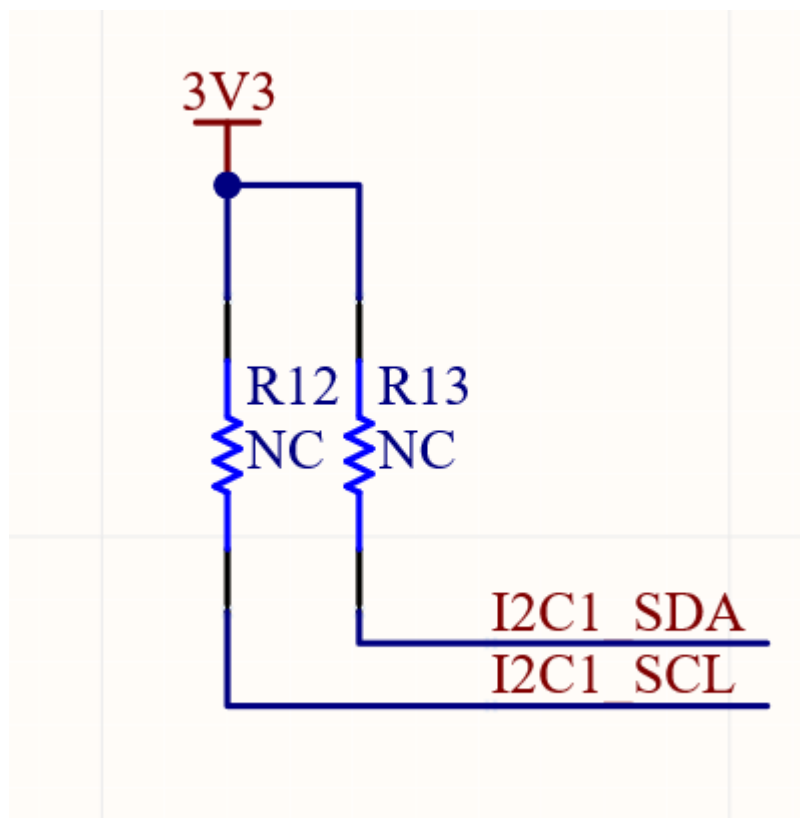


Figure 9: I2C Pull-up Resistance